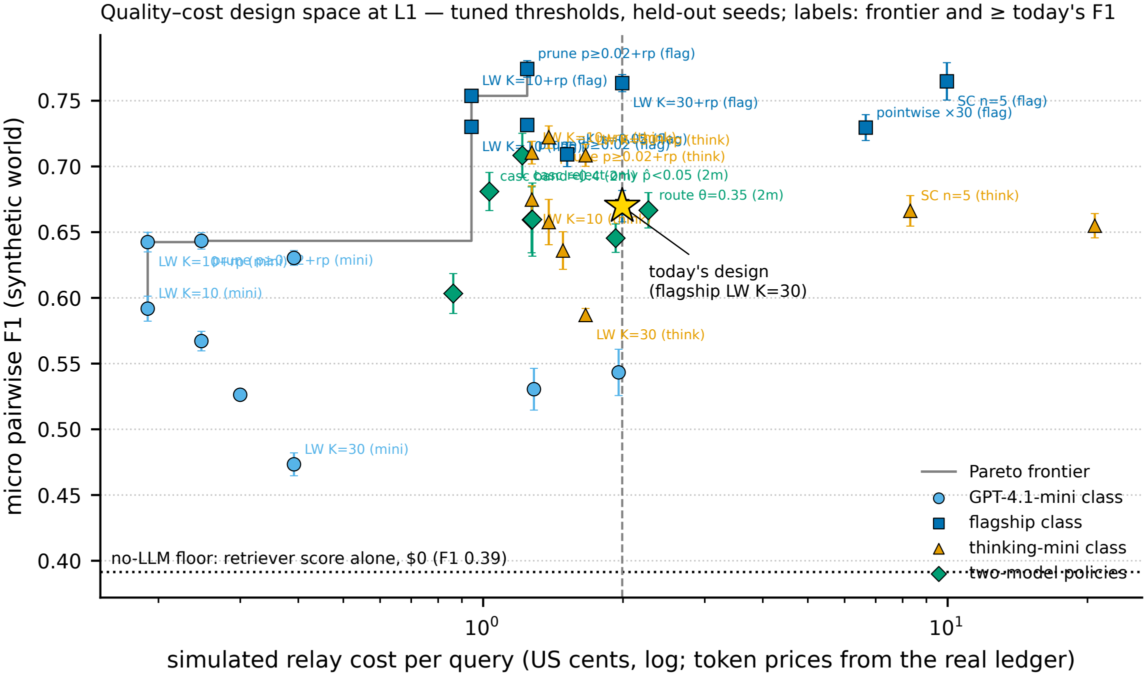
Quality–cost design space at L1 — tuned thresholds, held-out seeds; labels: frontier and ≥ today's F1
micro pairwise F1 (synthetic world)
0.75
0.70
0.65
0.60
0.55
0.50
0.45
0.40
simulated relay cost per query (US cents, log; token prices from the real ledger)
10^0
10^1
prune p≥0.02+rp (flag)
LW K=10+rp (flag)
LW K=30+rp (flag)
SC n=5 (flag)
pointwise ×30 (flag)
LW K=10 (flag)
prune p≥0.02 (flag)
LW K=10+rp (think)
prune p≥0.02+rp (think)
casc band=0.4 (2m)
reject-only p̂<0.05 (2m)
route θ=0.35 (2m)
LW K=10 (think)
SC n=5 (think)
LW K=30 (think)
today's design
(flagship LW K=30)
LW K=10+rp (mini)
prune p≥0.02+rp (mini)
LW K=10 (mini)
LW K=30 (mini)
no-LLM floor: retriever score alone, $0 (F1 0.39)
Pareto frontier
GPT-4.1-mini class
flagship class
thinking-mini class
two-model policies